

Merge Sort: Merging Small Piles of Playing Cards into Larger Sorted Piles

Computer scientist John von Neumann developed *merge sort* in 1945. It uses a divide-merge approach: each recursive call to `mergeSort()` divides the unsorted list into halves until they've been whittled down into lists of lengths of zero or one. Then, as the recursive calls return, these smaller lists are merged together into sorted order. When the last recursive call has returned, the entire list will have been sorted.

For example, the divide step takes a list, such as `[2, 9, 8, 5, 3, 4, 7, 6]`, and splits it into two lists, like `[2, 9, 8, 5]` and `[3, 4, 7, 6]`, to pass to two recursive function calls. At the base case, the lists have been divided into lists of zero or one item. A list with nothing or one item is naturally sorted. After the recursive calls return, the code merges these small, sorted lists together into larger sorted lists until finally the entire list is sorted. [Figure 5-3](#) shows an example using merge sort on playing cards.